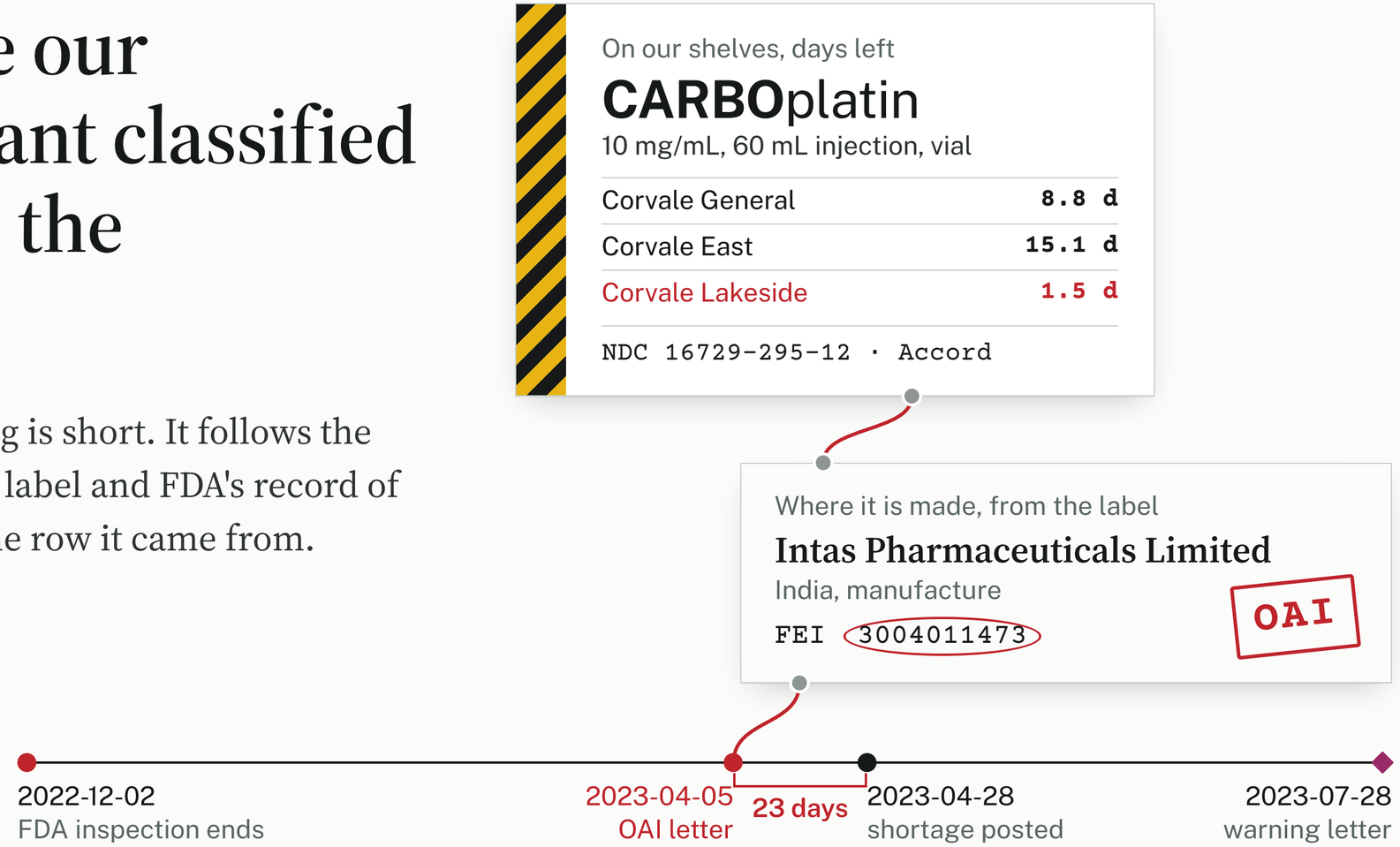
Upstream

FDA's records trace our carboplatin to a plant classified OAI 23 days before the shortage posting.

Upstream explains why a hospital drug is short. It follows the drug from the shelf to the plant on its label and FDA's record of that plant, and every sentence cites the row it came from.



A shortage arrives without its cause

20M hours

of staff time, nearly **$900M a year**, spent by US hospitals managing drug shortages.

Vizient survey, 2025

59%

of the shortages ASHP tracked in 2025 had a cause that was **unknown or not disclosed**.

ASHP data, reported by AJMC

227

active US drug shortages in Q2 2026.

ASHP, reported by AJMC

A hospital pharmacy buyer, Monday morning

"Carboplatin is short again. Why, and what do we do?"

Who we built for

The **pharmacy buyer** who has to answer that today, and the **pharmacist or director** who signs the decision. They piece it together by hand from FDA's website, wholesaler emails and their own stock spreadsheets. The facts are public, but scattered across a dozen FDA systems, keyed differently, and none of them knows what is on your shelf.

From the shelf to the plant, then back to a decision

One question runs nine fixed steps. The red part is FDA's record; the blue part is what Upstream writes, checks and hands to a person.



Shelf

Lakeside has 1.5 days of the 60 mL vial

NDC

`16729-295-12`, on FDA's list

Label

Accord's establishment block names the plant

Plant

Intas, India, `FEI 3004011473`

FDA record

OAI, letter dated 2023-04-05; import alert; warning letter

Cited why

6 sentences, each citing its rows, checked by a verifier

Options

Ranked by rule: transfer 3 vials from East to Lakeside

Memo

For the P&T committee, checked again

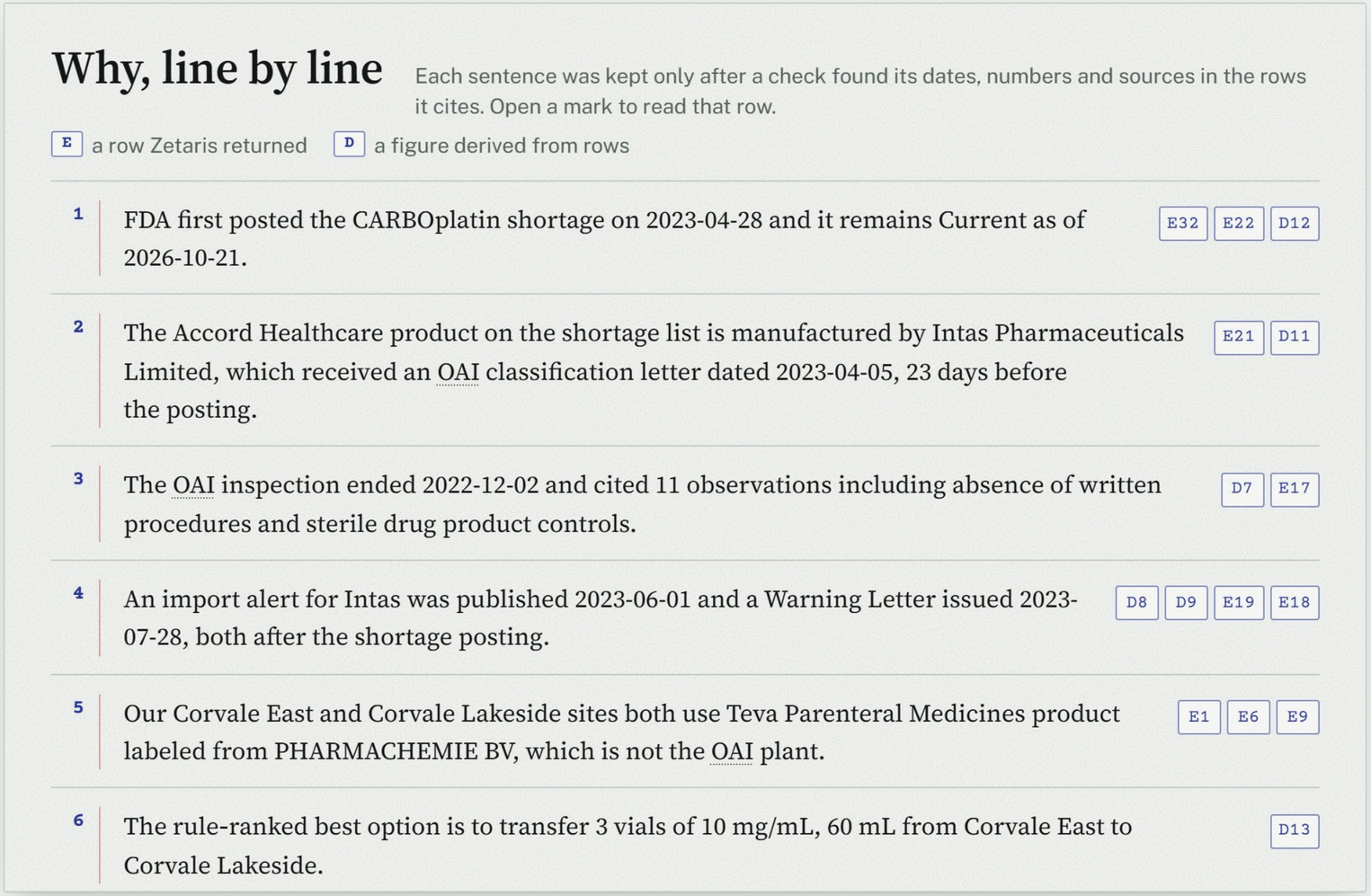
Approval

A named, signed-in pharmacist or the director

FDA found a problem — What Upstream wrote and verified

> "Our Accord carboplatin is made at Intas Pharmaceuticals' plant in Sanand, India (FEI 3004011473). FDA's inspection there ended 2022-12-02 and was classified OAI; the classification letter is dated 2023-04-05, 23 days before FDA posted the carboplatin shortage on 2023-04-28."
>
> The start of Upstream's answer. The run took 15 Zetaris statements across 13 systems in 24.7 s, and one model call.

A real answer, line by line



Why, line by line

Each sentence was kept only after a check found its dates, numbers and sources in the rows it cites. Open a mark to read that row.

E a row Zetaris returned D a figure derived from rows

1 FDA first posted the CARBOplatin shortage on 2023-04-28 and it remains Current as of 2026-10-21. E32 E22 D12

2 The Accord Healthcare product on the shortage list is manufactured by Intas Pharmaceuticals Limited, which received an OAI classification letter dated 2023-04-05, 23 days before the posting. E21 D11

3 The OAI inspection ended 2022-12-02 and cited 11 observations including absence of written procedures and sterile drug product controls. D7 E17

4 An import alert for Intas was published 2023-06-01 and a Warning Letter issued 2023-07-28, both after the shortage posting. D8 D9 E19 E18

5 Our Corvale East and Corvale Lakeside sites both use Teva Parenteral Medicines product labeled from PHARMACHEMIE BV, which is not the OAI plant. E1 E6 E9

6 The rule-ranked best option is to transfer 3 vials of 10 mg/mL, 60 mL from Corvale East to Corvale Lakeside. D13

Each sentence carries citation ids. A chip opens the Zetaris query and the row it came from.

`E17` a row Zetaris returned, here FDA inspection 1191818 at Intas

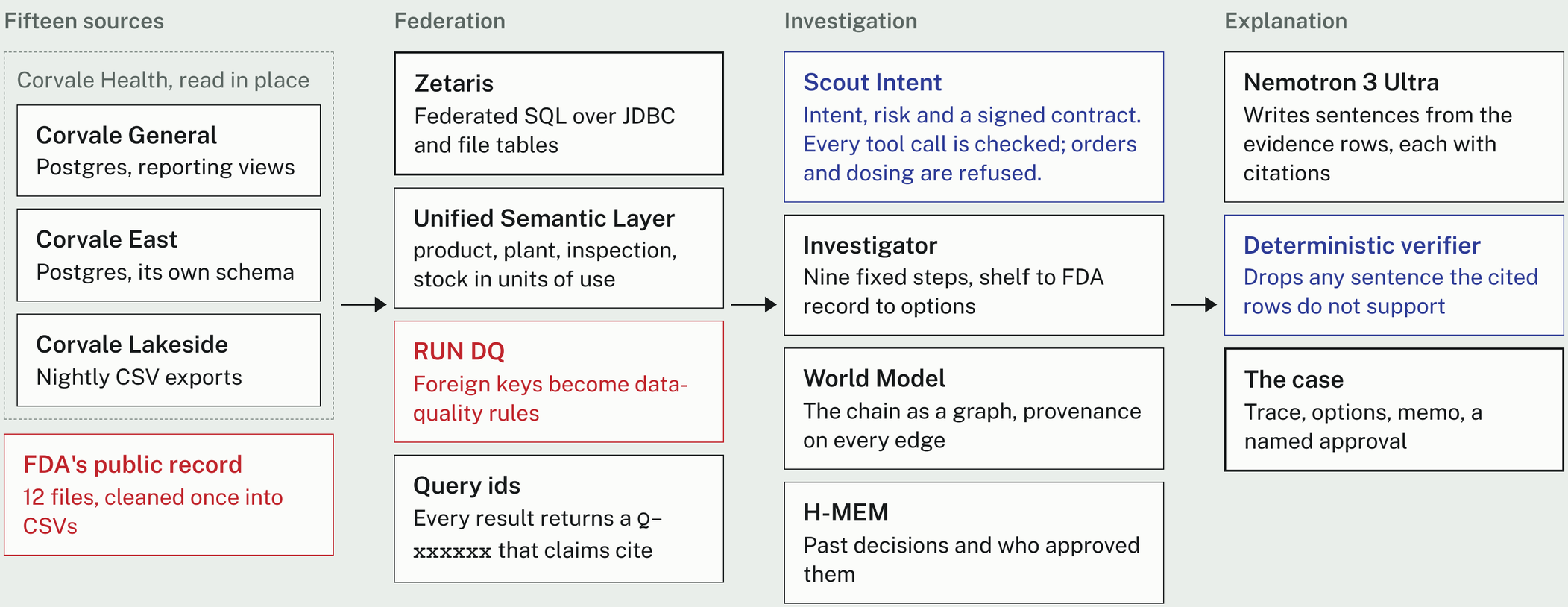
`D11` a figure derived from rows, here the 23-day gap

6 sentences and the headline checked against the record; 1 more was rejected and dropped.

15 Zetaris statements across 13 systems (3 hospital, 10 FDA files) in 24.7 seconds.

Live at `/case/carboplatin`, case `8dea84`, data as of 21 Oct 2026.

Architecture



Overnight: Markovian
FormularyWatch checks 215 drugs in 18 chunks; the cursor and findings live in World Model, not in the prompt.

Morning: Nemotron 3.5 Lightning
Writes the buyer's brief; a grounding check drops dates and numbers not in the watch rows.

Zetaris: hospitals read in place, FDA's record as tables

Federated, not copied. Two hospital Postgres servers over JDBC through a read-only account that sees only published views; one hospital's nightly CSV cabinet read as file tables where it lands.

FDA as file tables. 40,361 inspections, 46,893 Form 483 observations, 2,972 compliance actions, 1,007 import-alert listings, 12,341 recalls, plant lists from 1,532 label filings.

One semantic layer (`product, unit_of_use, site, site_fei, product_site, inspection`, a stock table per hospital). Lineage shows a stock query as "4 systems, 1 statement".

Units of use everywhere. East counts packages, so its activation multiplies by East's own pack size: a transfer is in real vials, per presentation, never pooled.

EXPLAIN shows the NDC list pushed to Postgres as `In(ndc11, [...])`, and a USL table as one opaque scan, so its filters are not pushed down. We report both.

RUN DQ on the semantic layer's keys

Rule	Fail	Query
Lakeside stock: NDC not on FDA's list	11 of 140	`Q-265662`
General and East stock: NDC and unit	0 of 686	`Q-02442a`
Label plant not in FDA's registration file	513 of 1,970	`Q-08b193`
Inspection FEI the file no longer holds	19,103 of 40,361	`Q-a61817`
Inspection id repeated	1,531 of 40,361	`Q-a61817`
Plant DUNS key in the registration file	0 of 14,793	`Q-5606ea`

An undashed NDC can be read three ways (4-4-2, 5-3-2, 5-4-1). The activation accepts a reading only when exactly one is on FDA's list; otherwise the row is marked ambiguous for the DQ rule. Seven Zetaris 2.4.3.1 lessons are written up in the README for other builders.

Meterless: all four engines, each doing real work

Engine	What it does in Upstream
Scout Intent	Decides intent, risk and scope; signs an HMAC execution contract. `assertContract()` runs before every tool. Shipped only as a spec, so we implemented it.
H-MEM	Remembers past decisions as the approver's quoted words, written only after a pharmacist or the director approves. `recallForDrug()`
World Model	Holds the supply chain as grounded entities with provenance; writes about unknown ids are refused.
Markovian	The overnight watch: 215 drugs in 18 chunks, with a fresh Scout contract per chunk.

Measured, 22 Oct 2026

13/13 · 8/8 · 8/8
H-MEM, World Model, Markovian on their unmodified conformance suites; Scout harness PASS

22/22
Block recall on 61 labelled pharmacy prompts, 15 held out: orders, dosing, injections always refused

60/60
Contract scope correct; tampered, wrong-secret and expired contracts refused

0.10 s
World Model, 300 drugs: one file write instead of 2,700 (29.2 s before our deferred flush)

A Scout harness PASS shows the plumbing is spec-shaped, not classification quality; that is what our own eval measures.

Intent accuracy 96% (51 of 53); injection recall 6 of 6; held-out block recall 4 of 4.

Every gated capability is refused until its approval exists; no contract is issued for a blocked prompt.

H-MEM recall with our pharmacy embedder: precision 0.938 against 0.125 for the reference as shipped (a calibration).

Markovian prompts stay at about 204 tokens per chunk at 300 drugs, because the state lives in World Model.

NVIDIA Nemotron writes; it never decides a fact

Nemotron 3 Ultra writes the explanation and the P&T memo. **Nemotron 3.5 Lightning** writes the overnight brief.

The state (confirmed, check, no signal) is set by rule, and every sentence must cite evidence ids.

A deterministic verifier checks dates, numbers, counts, stock fields, the event and direction of a day gap, plant and classification pairs, causes, status words and absence claims. A sentence that fails is dropped.

The memo and brief carry no citations, so they get a grounding check: over 214 recorded memos, **3,380 sentences checked, 38 dropped**.

44 of 44 false sentences written by three rounds of judges to slip past it were dropped; every true one tested was kept.

Some of them, as the verifier saw them

~~The OAI letter came 23 days after FDA posted the shortage.~~
the evidence says 23 days before, not after

~~The carboplatin shortage stems from the Intas OAI inspection 1191818.~~
causal claim ("stems from") the cited record does not make

~~No FDA action at Intas preceded the shortage posting.~~
false: the record holds an OAI inspection on 2022-12-02 at Intas

~~Corvale Lakeside has nine days of stock.~~
"nine" is a number in words; numbers must be digits from the evidence

It is pattern-based: a strong filter, not a proof. The test sentences are in `data/eval/verifier.json`.

Measured against FDA's raw files

The trace beats the model alone

59 plant links across 18 recent shortages; ground truth from separate Python over FDA's files that imports nothing from Upstream.

	Upstream	Nemotron Ultra, no tools
Reached the right plant	52 of 59	27 of 54
Right FDA FEI number	from the plant	0 of 54
Last inspection, class	56 of 59	30 of 54
Last inspection, date		0 of 54
Invented an inspection date	never	50 of 54
Said it was confident	n/a	54 of 54

The back-test: it explains, it does not forecast

Shortages FDA posted from 2022 with a placeable plant (92). Had FDA acted against a plant on the label in the two years before?



Strictly dated: **no lift** over products that never went short (1.1×, p 0.85).

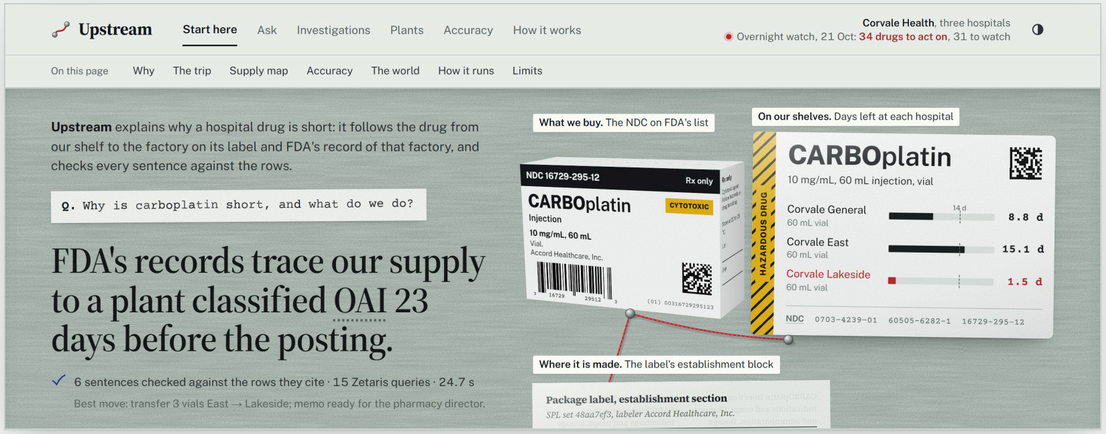
Close-dated: 1.93× the controls (p 0.012), but that uses hindsight about which inspections became OAI.

Counter-example: Baxter, Marion NC. The OAI was real; Hurricane Helene caused the 2024 IV fluid shortage.

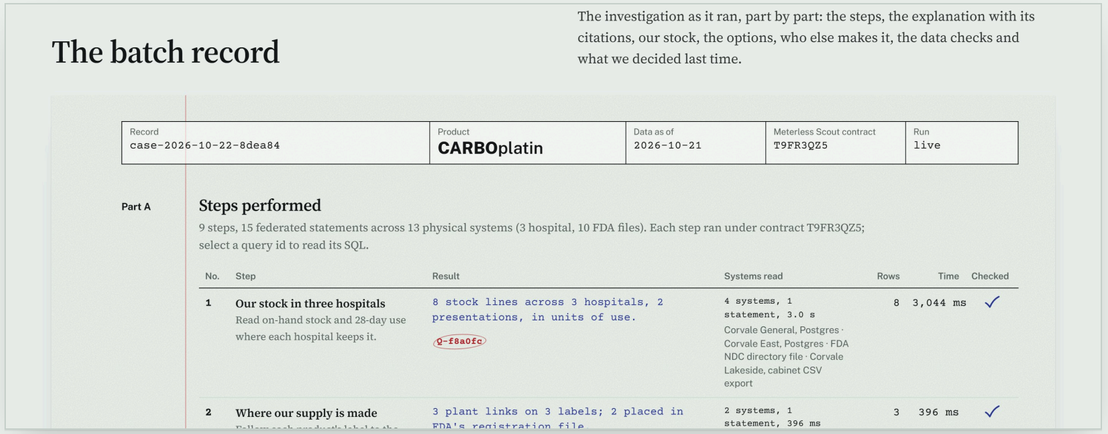
A plant's FDA record explains a shortage after the fact; it does not predict one. Upstream is built to explain, cite and act on what is known.

Where the trail exists, FDA acted about three months before the posting (median 98 to 103 days).

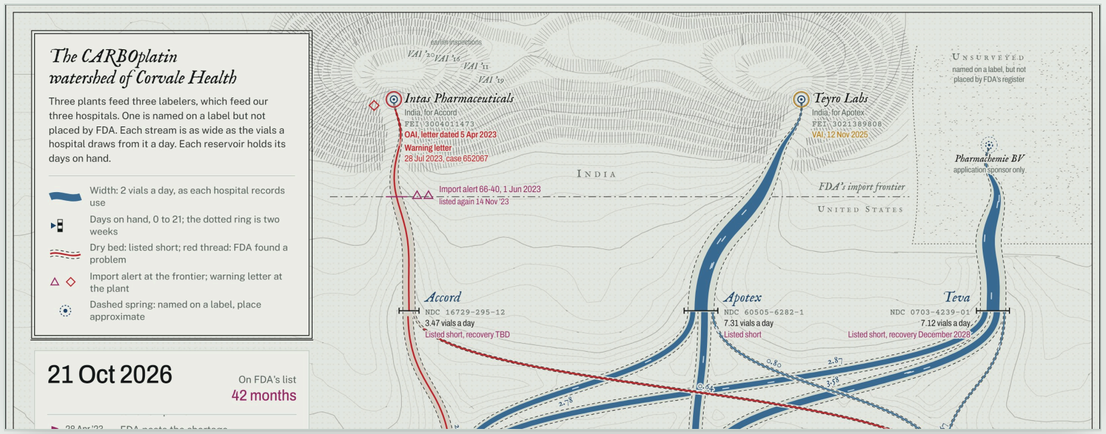
The product



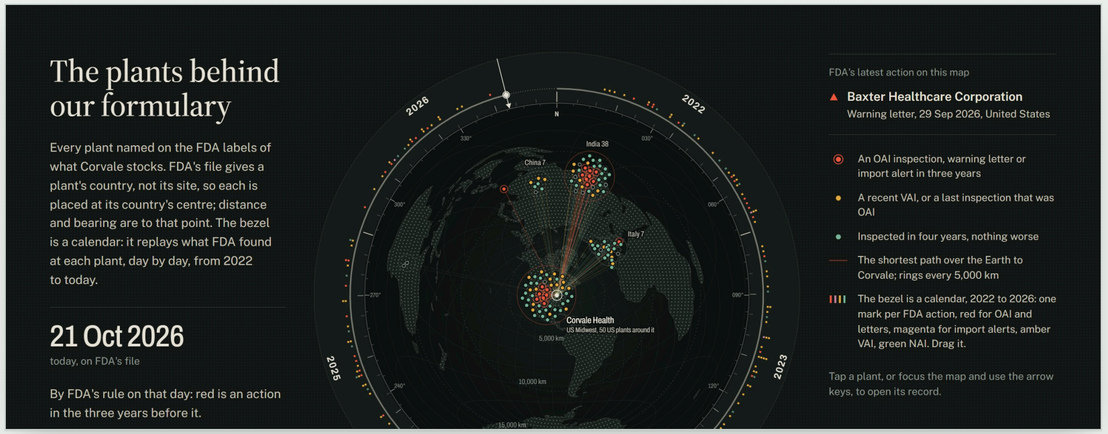
Home. The question, the cited answer, the thread to the plant.



The batch record. Every step, its query ids, systems and time.

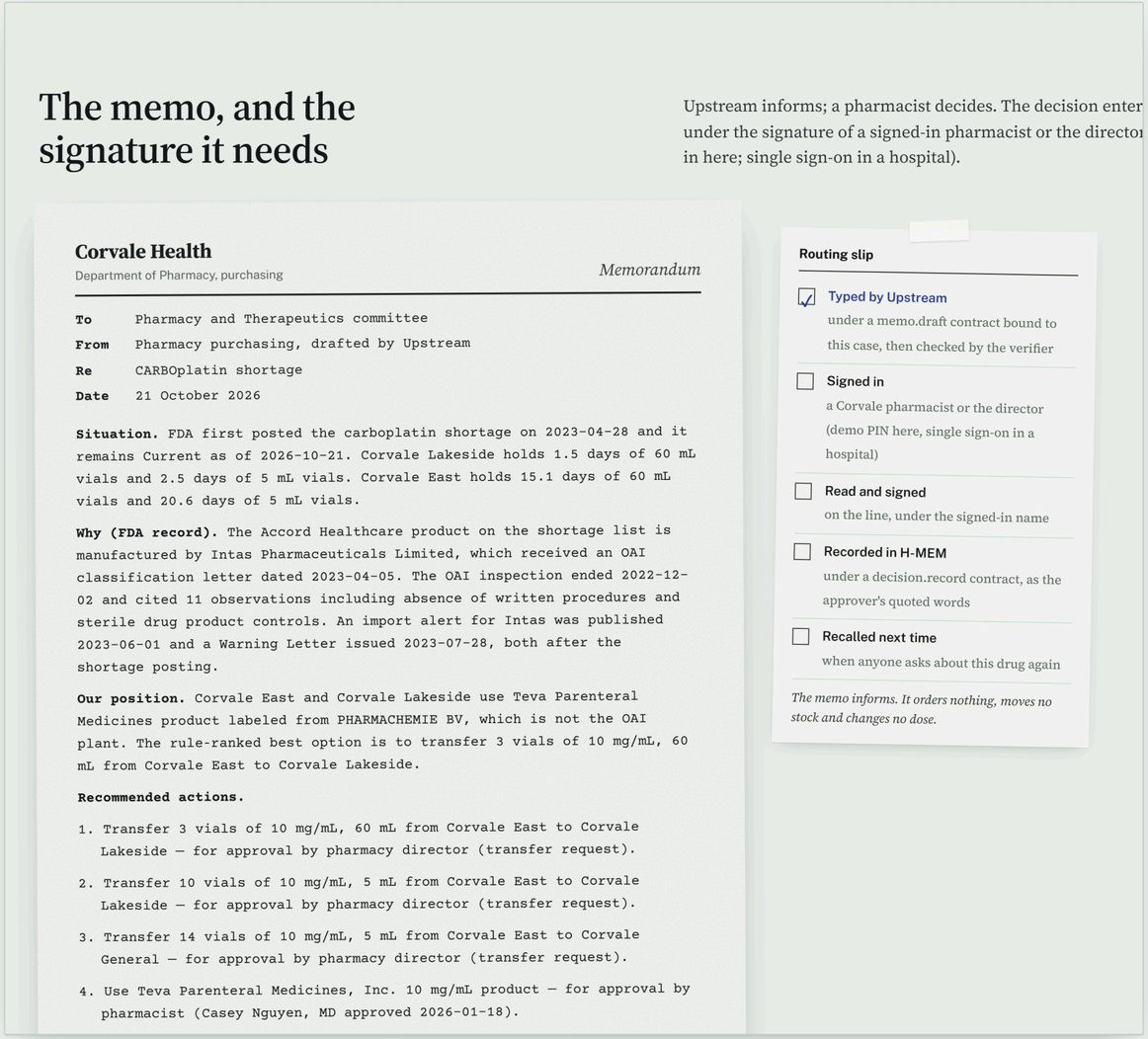


The watershed. Stream width is real daily use at each hospital.



The plants. The bezel is a calendar of FDA's actions.

Upstream informs; a pharmacist decides



A contract per question. Scout signs an execution contract; every tool call is checked against it before it runs. The carboplatin register: 11 entries, 11 allowed, 0 refused.

Refusals by policy. Orders, inventory changes, formulary changes and dosing advice are refused with a reason.

Named approval. Memory writes are refused until a pharmacist or the director approves. Name and role come from a session the server signs, never from the form; a buyer can draft, and Scout refuses the approval.

Single-use, short-lived. The memo's contract is bound to the case and drug and expires after five minutes. The memo itself goes through the verifier.

MCP server. Six read-only tools at `/api/mcp`; each checks its contract for every capability it reads.

Cursor built the hospitals; a pharmacist corrected them

Session 1, 11.5 minutes. Cursor's agent built Corvale Health's data generator from a written spec (`data/hospital/SPEC.md`): 14 file edits, 24 shell runs, the generator and its tests.

The review. A pharmacist found clinically wrong interchange pairs and wrote `FIXES-1.md`.

Session 2. Cursor applied the fixes (8 edits, 5 shell runs) until the free plan's usage limit stopped it. We finished the rest by hand; `audit.py` and 18 tests check them.

Both session logs are in `docs/cursor/`. The agent, the Zetaris setup, the engines and the site were written outside Cursor.

The stack

Zetaris 2.4.3.1, Docker on WSL2

Postgres ×2 and a CSV cabinet

Meterless engines, Apache-2.0, commit `1742dec`

NVIDIA Nemotron 3 Ultra, 3.5 Lightning

Next.js 15.5, React 19, TypeScript

Python 3.12: FDA bundle, generator, back-test

MCP, Streamable HTTP

Vercel, replaying recorded live runs

openFDA, DailyMed, FDA Data Dashboard

d3-geo for the plants map

What the pharmacist's review put into the product

Interchanges only a P&T committee could defend; cisplatin to carboplatin only "per treating oncologist".

Tall Man lettering (CARBOplatin) and ISMP strengths: mcg, never ug.

A plant whose last inspection was OAI is never shown as clear.

Transfers only within one presentation, in units of use, when the donor keeps two weeks.

What it does not do yet, and what comes next

Limits

Corvale Health is fictional. Its stock is synthetic; every NDC, labeler and plant is real FDA data.

Labels are incomplete. 23% name no plant and 33% list none in their establishment block; plants that make the active ingredient are almost invisible.

Look-ahead bias. Plant lists and import-alert lists are today's.

Explains, does not predict. The back-test shows no lift on strictly dated data.

The hosted site replays recorded live runs, and its approver sign-in is a demo.

Next

Real ERP and inventory connectors through Zetaris. The hospitals are already read in place, so a real system is a new data source, not a copy.

Per-tenant semantic-layer views, so each health system sees only its own stock.

Single sign-on in place of the demo PINs, feeding the same server-side approval rule.

More of FDA's record: drug master files and API plant registrations, to reach the ingredient makers.

Alerts from the overnight watch when FDA acts at a plant on our labels.

Every sentence cites its row.

Upstream traces a drug shortage from a hospital's shelf to the factory, and lets the pharmacist check every step.

Live site	upstream-shortage.vercel.app
Code	github.com/RohanGlitched/upstream
MCP	upstream-shortage.vercel.app/api/mcp

FDA inspection 1191818 at Intas ended 2022-12-02 and was classified OAI. E17

E17
FDA inspection 1191818
Intas Pharmaceuticals Limited
ended 2022-12-02
classified **OAI**